

Math 4441: Probability and Statistics

Prof. Muhammad Mahbub Alam, PhD

Department of Computer Science and Engineering

Islamic University of Technology

Chapter 3: Well-Known Random Variables and Distributions

(Part 1 – Discrete Well-known RVs)

Lectures 15 – 16

Math 4441: Probability and Statistics

References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Well-known discrete random variables

- Bernoulli random variable
- Geometric random variable
- Binomial random variable
- Pascal/Negative-binomial random variable
- Poisson random variable
- Hyper geometric random variable
- Discrete uniform random variable

Bernoulli random variable

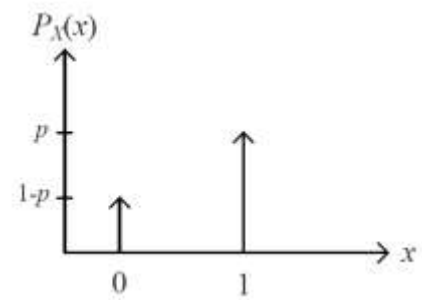
- An experiment with only two outcomes – success and failure - is defined as a Bernoulli experiment (BE)
- Even if an experiment has more than two outcomes, it can be converted into a Bernoulli experiment.
 - One of the outcomes is considered as a “success”, whereas all the other outcomes are considered as “failure” .
- RVs associated with a BE assumes the value 1 when a success occurs; and assumes a value 0, when a failure occurs,
 - Defined as a Bernoulli random variable
 - Bernoulli distribution

$P[X = 1]$ is the probability that a “success” occurs

$P[X = 0]$ is the probability that a “failure” occurs

Let p denote the probability that success occurs in a BE, where $0 < p < 1$.

$$P_X(x) = \begin{cases} 1 - p, & \text{for } x = 0, \\ p, & \text{for } x = 1, \\ 0, & \text{otherwise.} \end{cases}$$



$$F_X(x) = \begin{cases} 0, & \text{for } x < 0, \\ 1-p, & \text{for } 0 \leq x < 1, \\ 1, & \text{for } x \geq 1. \end{cases}$$

The Bernoulli random variable is the simplest, but nevertheless an important random variable

- Flip a coin; we may let
 - $s = \{head\} = \{success\}$
- Each computer chip produced is tested;
 - $s = \{the\ chip\ passes\ the\ test\}$
- A packet is transmitted;
 - $s = \{success\} = \{the\ packet\ is\ transmitted\ successfully\}$

Expected value and variance

$$\begin{aligned}E[X] &= \sum_{x=0}^1 xP_X(x) \\&= 0 \cdot (1-p) + 1 \cdot p \\&= p.\end{aligned}$$

$$\begin{aligned}Var[X] &= E[(X - E[X])^2] \\&= (0-p)^2 \cdot (1-p) + (1-p)^2 \cdot p \\&= p^2(1-p) + (1-p)^2p \\&= p(1-p)(p+1-p) \\&= p(1-p).\end{aligned}$$

Example: Pick a ball randomly from a box with 8 red balls and 7 blue balls. A success occurs when a red ball is picked.

Bernoulli Trials

BEs can be considered as the building block of many experiments. Such an experiment is often constituted by a number of successive BEs.

A sequence of BEs is called a Bernoulli trials, if

- Each BE is independent from the others
- Probability of success is a constant

1. Sending n data packets successively.
2. Sending a single packet successively until the transmission is successful.

1. Transmit a data packet from one station to another station. The sender continues to transmit the packet again and again until the transmission is successful. Assume the probability of success in each transmission is p , irrespective of the number of transmission already attempted for the packet. Therefore, this constitute a Bernoulli trials. The random variable associated with this Bernoulli trials might be the followings:
 - (a) The number of transmission required to transmit the data packet successfully.
 - (b) The number of unsuccessful transmissions before the packet is successfully sent.

2. Send n data packets from one station to another. Each of the packets is successfully sent with probability p , and it is dropped with probability $(1 - p)$, irrespective of whatever happened for the earlier packets. Assuming an attempt of successfully sending a packet or dropping it as an Bernoulli experiment, the above example of sending n packets becomes a Bernoulli trials. The following random variables can be defined for this Bernoulli trials:

- (a) The random variable X that denotes the number of packets successfully sent.
- (b) The random variable Y that denotes the number of packets dropped.

3. Consider the above example of sending n packets from one station to another with a little modification. Instead of sending n packets, the sending station continues to send packets one after another until r packets are successfully sent. The sequence of sending packets now becomes a Bernoulli trials, where the trial ends as soon as r packets are successfully sent. The random variables associated with the Bernoulli trials might include the followings:

- (a) The number of packets the sender attempt to send.
- (b) The number of packets dropped before the sender can send r packets.

In a Bernoulli trial, two quantities are important

- The number of repetitions of the BEs
- The number of successes

When one quantity is a constant, the other one is a variable; and becomes the random variable for the Bernoulli trial.

Well-known RVs associated with Bernoulli trials

	Number of repetitions	Number of successes	Random variables	Parameters	Distributions
Case 1					
Case 2					
Case 3					

PMFs of Well-known RVs associated with Bernoulli trials

Random Variable: X , PMF: $P_X(x)$, Probability of success in BE: p

Distributions	Para- meters	Number of success	Prob. of x success	Prob. of $n - x$ failures	# of ways	PMF
Binomial						
Geometric						
Pascal						

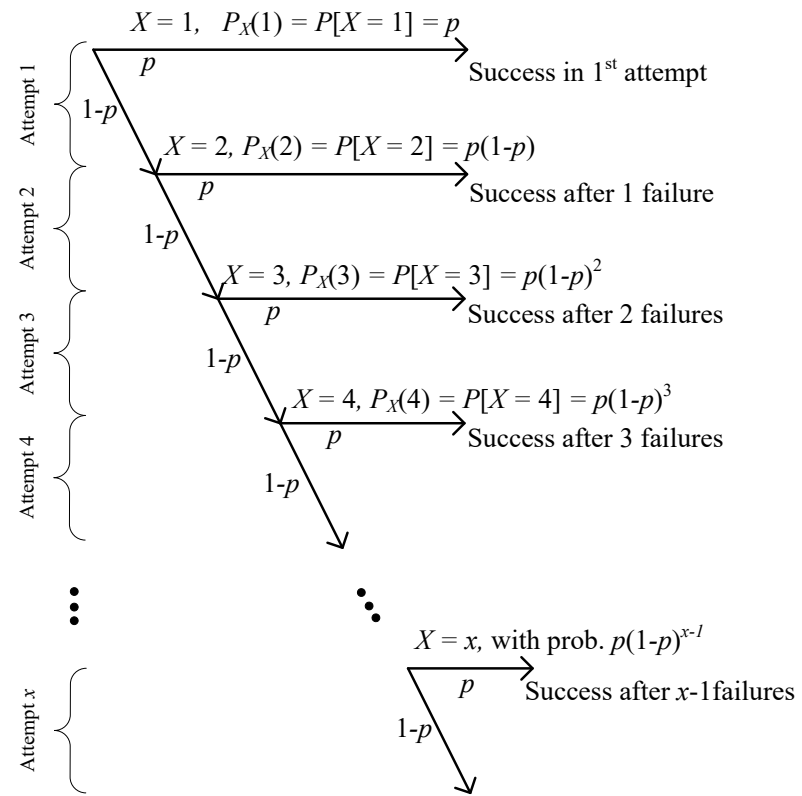
Geometric Distributions

Consider that a station in a wireless network transmits a data packet to another station. After transmitting the packet, the sending station waits for a predefined amount of time to receive feedback (i.e., an ACK) from the receiving station.

Let the random variable X denote the number of transmissions required to send the packet successfully. Assume that the probability of success of a packet is $\frac{1}{2}$.

$$S_X = \{1, 2, 3, \dots\}$$

x	1	2	3	4	...
$P[X = x]$	$\frac{1}{2}$	$(\frac{1}{2})^2$	$(\frac{1}{2})^3$	$(\frac{1}{2})^4$	...



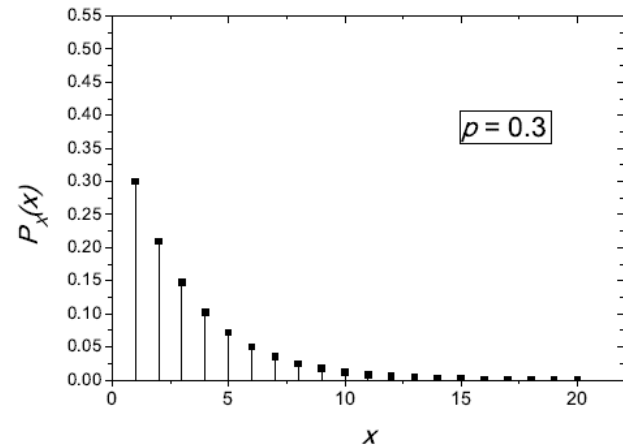
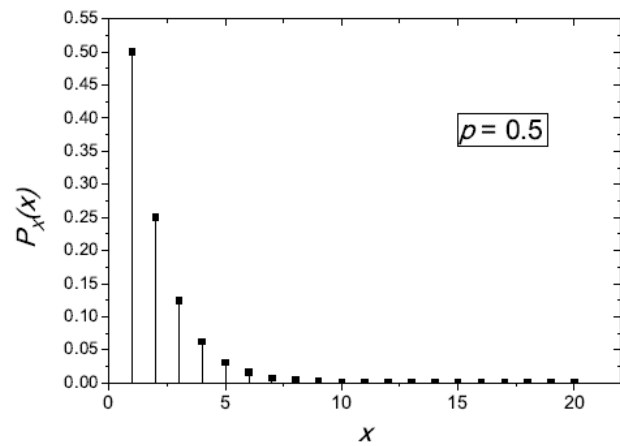
$$P[X = x] = \left(\frac{1}{2}\right)^x \quad \text{for } x = 1, 2, \dots$$

$$P[X = x] = 0 \quad \text{for } x \leq 0.$$

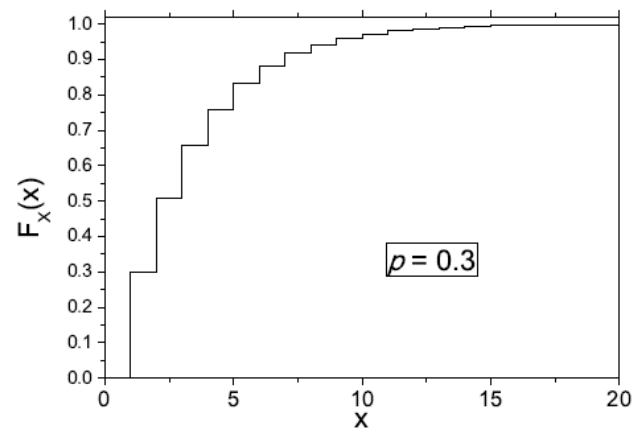
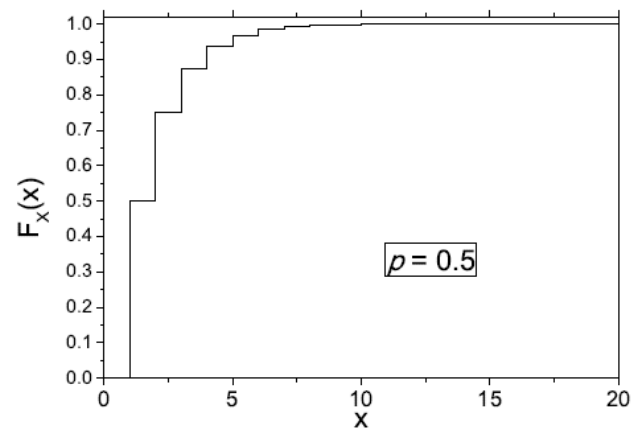
$$P_X(x) = \begin{cases} (\frac{1}{2})^x, & \text{for } x \geq 1; \\ 0, & \text{otherwise} \end{cases}$$

1. The trials are independent.
2. Each trial results in only two outcomes, labeled as “success” and “failure”.
3. The probability of a “success” in each trial, denoted as p , remains constant for the whole sequence.
4. The sequence of trials ends as soon as the first success occurs.

$$P_X(x) = \begin{cases} p(1-p)^{x-1}, & \text{for } x \geq 1; \\ 0, & \text{otherwise.} \end{cases}$$



$$F_X(x) = \begin{cases} 0, & \text{for } x < 1; \\ 1 - (1 - p)^x & \text{for } x \geq 1 \end{cases}$$



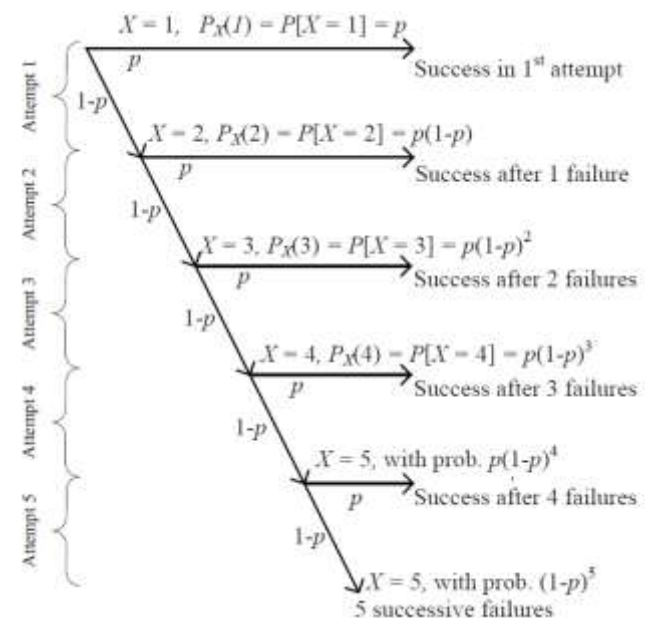
$$\begin{aligned}
E[X] &= \sum_{x=1}^{\infty} xP_X(x) \\
&= \sum_{x=1}^{\infty} x \cdot p(1-p)^{x-1} \\
&= p \sum_{x=1}^{\infty} \frac{d}{dq}(q^x) \quad [\text{where } q = 1-p] \\
&= p \frac{d}{dq} \left(\sum_{x=1}^{\infty} q^x \right) \\
&= p \frac{d}{dq} \left(\frac{q}{1-q} \right) \\
&= \frac{p}{(1-q)^2} = \frac{p}{p^2} \\
&= \frac{1}{p}.
\end{aligned}$$

$$\begin{aligned} \text{Var}[X] &= E[(X - E[X])^2] \\ &= \frac{(1-p)}{p^2}. \end{aligned}$$

Example: A father asks his sons to cut their backyard lawn. Since he does not specify which of the three sons is to do the job, each boy tosses a coin to determine the odd person, who must then cut the lawn. In the case that all three get heads or tails, they continue tossing until they reach a decision. Let p be the probability of heads and $q = 1 - p$, the probability of tails.

- a) Find the PMF of the number of rounds (tosses) required.
- b) Find the expected number of rounds required.

Truncated Geometric Distribution

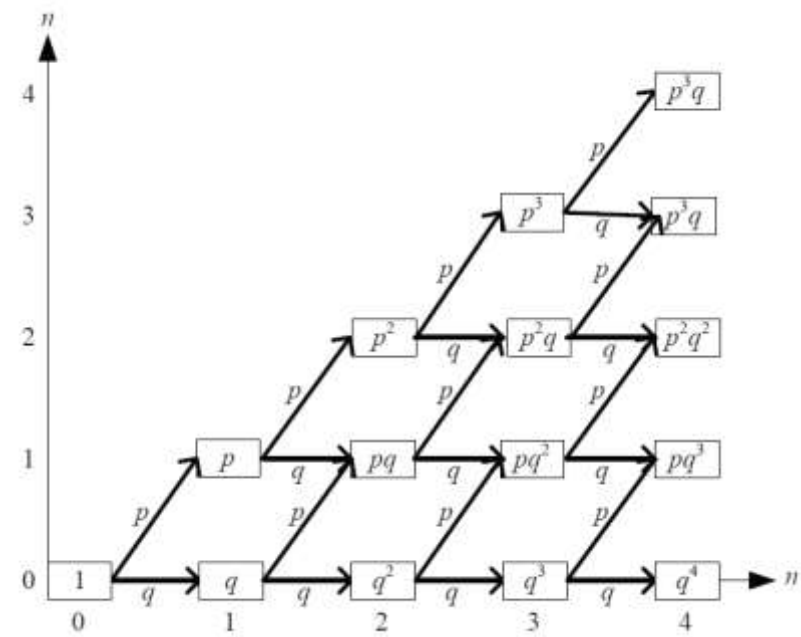


$$P_X(x) = \begin{cases} p(1-p)^{x-1}, & \text{for } x = 1, 2, \dots, R-1; \\ (1-p)^{R-1}, & \text{for } x = R; \\ 0, & \text{otherwise.} \end{cases}$$

Binomial Distribution

In a Bernoulli trial, if

- Number of repetitions is fixed
- Number of success(es) is a variable and is defined as the RV
- The RV is called a binomial RV and its family of distribution is called binomial family of distribution



$$P_X(k) = \begin{cases} \binom{n}{k} p^k q^{n-k}, & \text{for } k = 0, 1, 2, \dots; \\ 0, & \text{otherwise.} \end{cases}$$

The CDF of a binomial RV is

$$F_X(x) = \sum_{k=0}^x \binom{n}{k} p^k q^{n-k}$$

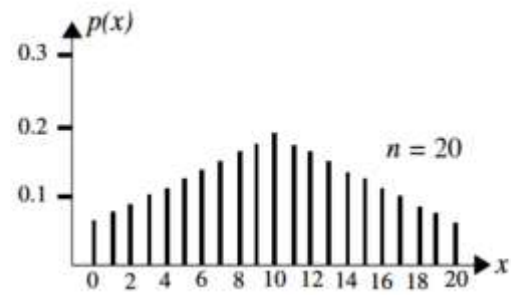
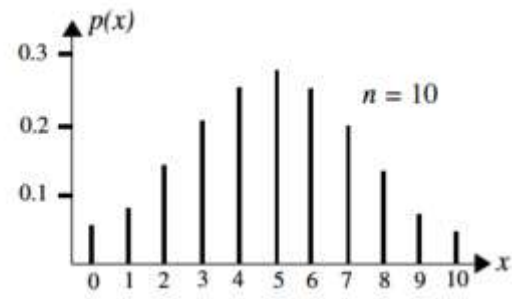
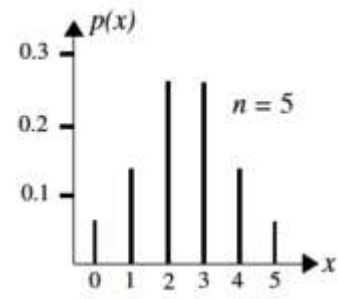
$$E(X) = np, \quad \text{Var}(X) = np(1-p), \quad \sigma_X = \sqrt{np(1-p)}.$$

Why is it called binomial RV/Distribution

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

is the binomial coefficient in the binomial theorem

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$



$$\frac{P_X(k)}{P_X(k-1)} = \frac{(n-k+1)p}{kq} = 1 + \frac{(n+1)p-k}{kq}.$$

Negative Binomial Distribution

In a Bernoulli trial, if

- Number of success(es) is fixed
- Number of repetitions is a variable and is defined as the RV
- The RV is called a negative binomial or Pascal RV and its family of distribution is called Pascal family of distribution

PMF of Pascal RVs

In order to determine $P_X(n)$ for this case, let A be the event that the first $n - 1$ trials yield exactly $k - 1$ successes, regardless of their order, and B be the event that a success turns up at the n th trial. Then, owing to independence,

$$P_X(n) = P[A \cap B] = P[A]P[B].$$

Now, $P[A]$ obeys a binomial distribution with parameters $n - 1$ and $k - 1$, or

$$P[A] = \binom{n-1}{k-1} p^{k-1} q^{n-k}, \quad n = k, k+1, \dots,$$

and $P[B]$ is simply

$$P[B] = p.$$

Combining the above three equations, we have

$$P_X(n) = \binom{n-1}{k-1} p^k q^{n-k}, \quad n = k, k+1, \dots$$

Hypergeometric Distribution

In a box, there are r red and g green balls. Pick n balls randomly

(a) with replacement and

(b) without replacement.

Find the probability that exactly X red balls are picked. ($x \leq \min(r, n)$ and $n \leq r + g$ for without replacement.)

$$P_X(x) = \begin{cases} \frac{\binom{r}{x} \binom{g}{n-x}}{\binom{r+g}{n}}, & x = 0, 1, \dots, \min(r, n) \\ 0, & \text{otherwise.} \end{cases}$$

$$E[X] = \frac{nr}{r+g}$$

$$V[X] = \frac{n(rg)}{(r+g)^2} \left(1 - \frac{n-1}{r+g-1} \right)$$

Poisson Random Variable

$$P_X(x) = \begin{cases} \frac{(\lambda T)^x e^{-(\lambda T)}}{x!}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

$$E[X] = \lambda T$$

$$V[X] = \lambda T$$

$$P_X(x) = \begin{cases} \left(\frac{e^{-\alpha} \alpha^x}{x!} \right), & \text{for } x = 0, 1, 2, \dots; \\ 0, & \text{otherwise,} \end{cases}$$

Example 2.8. The number of wrong numbered calls that arrive in a certain office has the Poisson distribution with the rate $\lambda = 2$ calls per day. Find the probability that there will be at least 2 wrong calls by tomorrow.

$$\lambda = 4 \text{ calls/day}$$

$$T = 2 \text{ days}$$

$$\alpha = \lambda T = 4 \times 2 = 8 \text{ calls}$$

$$p = P[X \geq 2]$$

$$= 1 - P[X < 2]$$

$$= 1 - P[X = 0] - P[X = 1]$$

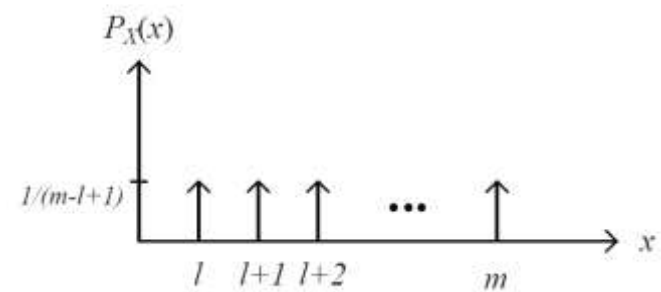
$$= 1 - \frac{e^8 \times 8^0}{0!} - \frac{e^8 \times 8^1}{1!}$$

$$= 1 - e^8 - 8e^8$$

$$= 1 - 9e^8.$$

Uniform Random Variable

$$P_X(x) = \begin{cases} \frac{1}{m-l+1}, & \text{for } x = l, l+1, l+2, \dots, m; \\ 0, & \text{otherwise.} \end{cases}$$



$$F_X(x) = \begin{cases} 0, & \text{for } x < l, \\ \frac{x-l+1}{m-l+1}, & \text{for } x = l, l+1, \dots, m-1, \\ 1, & \text{for } x \geq m. \end{cases}$$

$$E[X] = \frac{m+l}{2}$$

$$V[X] = \frac{(m-l)(m-l+2)}{12}$$

Quiz 1

A box (Box *A*) contains 8 red balls and 5 blue balls. Another box (Box *B*) contains 4 red balls and 7 blue balls. A ball is drawn (first ball) from the Box *A* and is transferred to Box *B*. Another ball is then drawn (second ball) from Box *B* and is transferred to Box *A*.

1.

Define appropriate events and calculate the probability that the color of the first ball was red, if it is given that the color of the second ball is red.

6

Assume another ball is drawn (third ball) from Box A and is transferred to Box B.

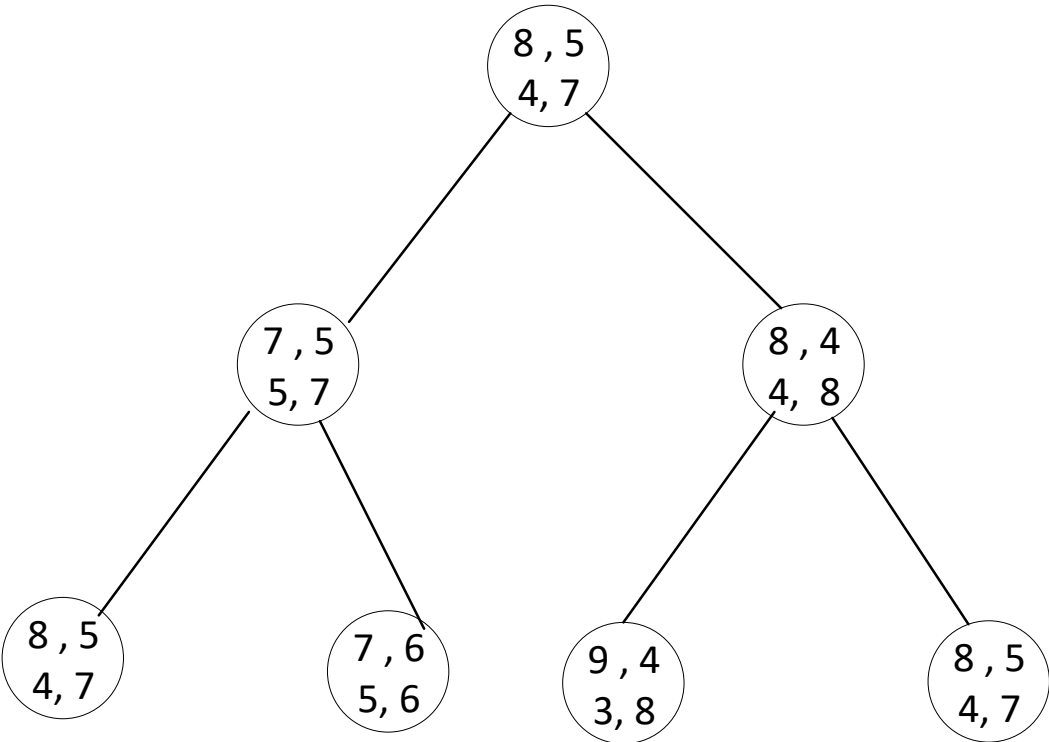
2.

Define appropriate events and calculate the probability that the color of the third ball is red.

5
3.

If the color of the third ball is read, calculate the probability that the color of the first ball and the second ball were red.

4



Quiz 2:

1. The probability density function of a random variable X is given by

6

$$f(x) = \begin{cases} c(x-1)(2-x), & 1 < x < 2 \\ 0, & \text{Otherwise.} \end{cases}$$

Calculate the value of c and find the cumulative distribution function (CDF) of X .

2. A 4-sided fair die is tossed twice. The sum of the outcomes is denoted by X and the largest value by Y .

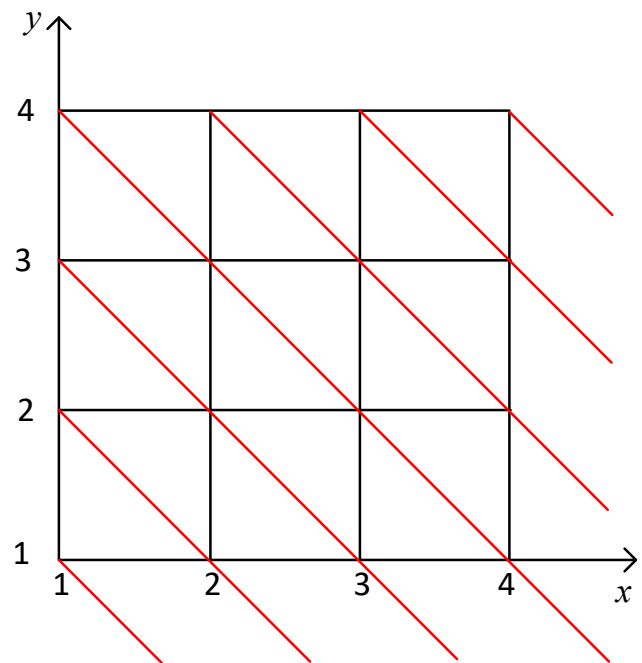
- a) Calculate the joint probability mass function of X and Y . (Show the detailed calculation of the joint PMF of each of the (x, y) pairs with non-zero probabilities.)
- b) Find the marginal probability mass functions of X and Y .

7

2

$$\int_1^2 c(x-1)(2-x) dx = 1 \implies c \left[-\frac{x^3}{3} + \frac{3x^2}{2} - 2x \right]_1^2 = 1 \implies c = 6$$

$$F(x) = \begin{cases} 0 & x < 1 \\ -2x^3 + 9x^2 - 12x + 5 & 1 \leq x < 2 \\ 1 & x \geq 2. \end{cases}$$



	$Y = 1$	$Y = 2$	$Y = 3$	$Y = 4$	
$X = 2$	1/16				
$X = 3$		2/16			
$X = 4$		1/16	2/16		
$X = 5$			2/16	2/16	
$X = 6$			1/16	2/16	
$X = 7$				2/16	
$X = 8$				1/16	